



DOC ID: SL-POP-ERP-MS-002

BA Reference: DOC-002 v1.0 • Enterprise Edition

POPULAR AUTO SPARE & A/C PARTS CO. W.L.L

MODULE 02 OF 09

PROCUREMENT & VENDOR PORTAL

Module 2: Vendor & Purchase Lifecycle, Supplier Portal & 3-Way Matching

Comprehensive enterprise procurement architecture, dedicated engineering team rate matrix, 12-week deliverable breakdown, and milestone payment agreement for Popular Auto Spare & A/C Parts Co. W.L.L.

TOTAL DURATION

12 Weeks

CALENDAR SPAN

3.00 Months

TOTAL INVESTMENT

BD 4,570.909

PAYMENT MODEL

4 Gates (25%)

01

Section 1.0: Executive Summary & Multi-Module Roadmap Overview

MASTER CONTEXT

Module 02: Vendor & Purchase Management, Supplier Portal & 3-Way Matching governs the end-to-end multi-country procurement lifecycle across Bahrain and Dubai operations. Synthesized directly from the **DOC-002 v1.0 Business Analysis Document**, it transforms manual phone/email purchasing into an automated 15-step digital pipeline: from consolidated branch stock shortages ("Less Item List") and CTO strategic approval, to automated multi-vendor RFQ bidding, landed cost currency tracking, Physical Verification Note (PVN) barcode intake, and automated 3-way invoice matching.

 **Complete 9-Module ERP Implementation Architecture (Module 02 Highlighted in Blue)**

MODULE 01

PCode & Item Master Engine

FIFO sequential PCode numbering, 4-stage CTO approval, temporary item hold, cross-reference mapping, PartSouq API & multi-tier pricing.

10 Wks • 4 Gates

BD 3,809.091

MODULE 02 • CURRENT FOCUS

Vendor & Purchase Lifecycle

Global supplier cataloguing, 3-way RFQ procurement, landed cost tracking, custom clearance & automated replenishment.

12 Wks • 4 Gates

BD 4,570.909

MODULE 03

Store Verification & Stock Control

Inward store verification (PVN), blind 60/40 cycle counts, 5-tier bin numbering, 7-stage GIT transfers & quarantine management.

15 Wks • 4 Gates

BD 5,713.636

MODULE 04

Multi-Branch Sales & POS Engine

Multi-channel POS, customer credit limits, multi-tier discount approval, quotation pipeline & return management.

15 Wks • 4 Gates

BD 5,713.636

MODULE 05

Financial Accounting & VAT

Multi-currency general ledger, AP/AR automation, National Bureau for Revenue (NBR) VAT returns & audit trials.

16 Wks • 4 Gates

BD 6,094.545

MODULE 06

Enterprise Administration & Roles

Centralized multi-company configuration, RBAC, division isolation, system audit vault & automated backups.

14 Wks • 4 Gates

BD 5,332.727

MODULE 07

Human Resource Management

Bahrain LMRA/SIO compliance, automated WPS payroll processing, leave accrual, document expiry & EOSB calculations.

10 Wks • 4 Gates

BD 3,809.091

MODULE 08

Hardware & Infrastructure Setup

Local branch failover server architecture, thermal barcode printer mesh, high-speed PDA Wi-Fi roaming & UPS power backup.

08 Wks • 4 Gates

BD 3,047.273

MODULE 09

Executive BI & Analytics Dashboard

Real-time GM executive command center, branch turnover analysis, aging inventory heatmap & AI-driven predictive replenishment.

08 Wks • 4 Gates

BD 3,047.273

02

Section 2.0: Comprehensive Scope Matrix & Technical Architecture

DOC-002 REQUIREMENTS

Synthesized directly from the **DOC-002 v1.0 Business Analysis Document**, Module 2 replaces fragmented purchasing with a structured 15-step governance workflow connecting branch managers, central procurement, CTO executive approvals, external suppliers, and main warehouse receivers.



Supplier Master & Communication Portal

Centralized supplier repository with transaction-specific magic links and secure chat communication, ensuring complete permanent audit history independent of individual employee turnover.



Consolidated "Less Item List" Engine

Automated pooling engine that checks main warehouse stock for branch requisitions and aggregates unfulfilled items into a centralized shortage pool for bulk procurement negotiations.



Pattern Analysis & CTO Strategic Panel

Historical consumption and seasonal purchase pattern intelligence paired with an executive control dashboard empowering the CTO to review, adjust, and authorize purchase requests.



Automated RFQ & 3-Way Vendor Ranking

Single-click multi-supplier RFQ generation (web portal + Excel/PDF export) with real-time quote submission and side-by-side comparative ranking based on price, landed cost, and lead time.



Multi-Vendor Split PO & Landed Cost

One-click multi-vendor split Purchase Order generation with multi-currency exchange rate lock (USD/EUR/JPY/AED/BHD) and automated landed cost/customs duty allocation across items.



PVN & Strict 3-Way Invoice Matching

Physical Verification Note (PVN) barcode scanning intake at main store, strict 3-way matching (PO vs PVN vs Supplier Invoice) with over-receipt protection, and automatic global stock updates.

03

Section 3.0: Dedicated Engineering Team Cost & Resource Allocation Matrix

TRANSPARENT RATES

Pricing for Module 2 is determined directly by allocating a 5-person dedicated engineering team over **12 working weeks (3.00 calendar months)**. Rates are computed on standard 160-hour monthly engineering allocations in Bahraini Dinars (BHD):

Project Manager & Solutions Architect

BD 545.455

BD 136.364 / Wk • 3.00 Mo = BD 1,636.365

100% Dedicated

Back-End Lead Engineer (PHP MVC / APIs)

BD 227.273

BD 56.818 / Wk • 3.00 Mo = BD 681.819

100% Dedicated

Front-End Lead Engineer (React 18 / UI)

BD 227.273

BD 56.818 / Wk • 3.00 Mo = BD 681.819

100% Dedicated

Database & Cloud DevOps Architect

BD 204.545

BD 51.136 / Wk • 3.00 Mo = BD 613.635

100% Dedicated

QA & System Test Automation Lead

BD 159.091

BD 39.773 / Wk • 3.00 Mo = BD 477.271

100% Dedicated

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	3.00 Months (12 Wks)	BD 1,636.365
Back-End Lead Engineer (PHP MVC / REST APIs / Procurement)	BD 227.273	BD 56.818	BD 1.420	3.00 Months (12 Wks)	BD 681.819
TOTAL MODULE 2 DEDICATED ENGINEERING INVESTMENT				12 Working Weeks	BD 4,570.909

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H)	EFFORT ALLOCATION	TOTAL COST (BHD)
Front-End Lead Engineer (React 18 / Supplier Portal UI)	BD 227.273	BD 56.818	BD 1.420	3.00 Months (12 Wks)	BD 681.819
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.00 Months (12 Wks)	BD 613.635
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	3.00 Months (12 Wks)	BD 477.271
TOTAL MODULE 2 DEDICATED ENGINEERING INVESTMENT				12 Working Weeks	BD 4,570.909

04

Section 4.0: Detailed Milestone Breakdown & Deliverables (12-Week Roadmap)

4 DELIVERABLE GATES @ 25%

Module 2 delivery is structured across 4 sequential, independently verifiable engineering milestone tranches over a 12-week implementation lifecycle. Each milestone requires technical demonstration and UAT sign-off before gate payment authorization:

Milestone 1: Supplier Master Database, Communication Portal & "Less Item List" Engine

🕒 Weeks 1-3 • 25% (BD 1,142.727)

Builds the centralized supplier directory for local and international vendors, deploys secure transaction magic-link communication interfaces with permanent chat logs, and implements the branch requisition checking and consolidated "Less Item List" shortage pooling engine.

☰ **KEY ENGINEERING DELIVERABLES:**

- ✓ Supplier Master entity architecture with multi-country categorization, payment terms, and currency settings.
- ✓ Supplier Communication Portal: Secure magic link access with integrated transaction chat and file exchange.
- ✓ Branch Requirement to Main Store Availability Check workflow and Store Delivery Note generation.
- ✓ Centralized "Less Item List" aggregation engine automatically compiling unfulfilled branch stock requests.
- ✓ Automated Low-Stock & Reorder Level monitoring dashboards with branch consumption velocity metrics.

Acceptance Verification: Branch requisition routing to main store verified; supplier magic-link chat operational; "Less Item List" consolidation tested with 1,000 branch request lines.

Gate Payment: BD 1,142.727

Milestone 2: Purchase Pattern Analysis, CTO Strategic Approval Gate & Multi-Vendor RFQ Engine

🕒 Weeks 4-6 • 25% (BD 1,142.727)

Implements historical purchase pattern intelligence, CTO executive approval dashboard with quantity adjustments, automated multi-vendor RFQ generation (portal links + Excel/PDF export), and automated 3-way vendor quotation ranking matrix.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ Purchase Pattern Analytics Engine analyzing historical pricing, seasonal demand, and vendor lead times.
- ✓ CTO Strategic Control Panel for purchase request approval, quantity modifications, and budget authorization.
- ✓ Automated Multi-Vendor RFQ Dispatch Engine (batch dispatch via portal links and standardized Excel templates).
- ✓ Vendor Quotation Submission Portal with line-item pricing, availability dates, and brand specification entry.
- ✓ Automated Vendor Ranking Matrix: 3-way comparative evaluation (price, landed cost, lead time, and payment terms).

Acceptance Verification: Multi-vendor RFQ batch dispatch tested across 5 suppliers; real-time quote submission and side-by-side automated comparative ranking matrix validated.

Gate Payment: BD 1,142.727

Milestone 3: Multi-Vendor Split PO Generation, Landed Cost Tracking & Exchange Rate Lock

🕒 Weeks 7-9 • 25% (BD 1,142.727)

Deploys one-click multi-vendor split Purchase Order generation from evaluated quotes, integrates multi-currency exchange rate locks (USD, EUR, JPY, AED, BHD), international landed cost calculation, and shipment milestone tracking.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ Automated Purchase Order (PO) Issuance Engine supporting multi-vendor split generation from single RFQs.
- ✓ Multi-Currency Exchange Rate Lock Engine with real-time conversion and financial commitment tracking.
- ✓ International Landed Cost Engine prorating customs duties, freight, insurance, and handling charges per item.
- ✓ Supplier Order Confirmation, Acknowledgment, and Shipment Tracking portal with ETA milestone alerts.
- ✓ Purchase Order Amendment & Cancellation governance workflows with CTO re-authorization triggers.

Acceptance Verification: Split PO generation verified across 3 winning vendors; landed cost calculation verified against customs formula; currency lock functionality verified.

Gate Payment: BD 1,142.727

Milestone 4: Physical Verification Note (PVN), 3-Way Invoice Matching & Final UAT Sign-Off

🕒 Weeks 10–12 • 25% (BD 1,142.727)

Implements Main Store Physical Verification Note (PVN) barcode scanning intake, automated 3-way invoice matching (PO vs PVN vs Invoice), partial delivery backorder management, global stock ledger sync, and comprehensive stakeholder UAT sign-off.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ Purchase Department Commercial Invoice Verification and preliminary pricing audit staging screen.
- ✓ Main Store Physical Verification Note (PVN) barcode scanning module with damage and discrepancy flagging.
- ✓ Strict 3-Way Matching Engine (PO vs PVN Physical Intake vs Supplier Commercial Invoice) with over-receipt lock.
- ✓ Partial Shipment & Backorder Management Engine with automated pending quantity tracking.
- ✓ Final Purchase Confirmation, Automatic Stock Ledger Updates, Multi-Branch Visibility & Final Client UAT Sign-Off.

Acceptance Verification: 3-way matching blocks discrepancies outside tolerance; PVN barcode scan operational; 100% test scenario pass rate in formal stakeholder UAT.

Gate Payment: BD 1,142.727

05

Section 5.0: Milestone Payment Structure & Invoicing Schedule (12-Week Model)

GUARANTEED MILESTONE GATES

Payments are strictly gated to verified milestone delivery and technical demonstration sign-off across 4 equal tranches of 25%:

INVOICE TRANCHE	TARGET TIMELINE	MILESTONE DELIVERABLE SCOPE	% VALUE	NET AMOUNT (BHD)	VAT 10% (BHD)	GROSS TOTAL (BHD)
Tranche 1 (Gate 1)	Week 3 Completion	Supplier Master Database, Communication Portal & Less Item List Engine	25.00%	BD 1,142.727	BD 114.273	BD 1,257.000
Tranche 2 (Gate 2)	Week 6 Completion	Purchase Pattern Analysis, CTO Strategic Approval Gate & RFQ Engine	25.00%	BD 1,142.727	BD 114.273	BD 1,257.000
Tranche 3 (Gate 3)	Week 9 Completion	Multi-Vendor Split PO Generation, Landed Cost Tracking & Currency Lock	25.00%	BD 1,142.727	BD 114.273	BD 1,257.000
Tranche 4 (Gate 4)	Week 12 Completion	Physical Verification Note (PVN), 3-Way Invoice Matching & Final UAT	25.00%	BD 1,142.727	BD 114.273	BD 1,257.000
TOTAL MODULE 2 INVESTMENT (12 WORKING WEEKS)			100.00%	BD 4,570.909	BD 457.091	BD 5,028.000

06

Section 6.0: Working Calendar, SLA, 15-Day Grace Period & Legal Terms

STANDARD SLA

1. 15-Day UAT Review SLA & Automatic Acceptance Grace Period

- Upon delivery of each milestone build to the staging environment, Popular Auto Spare will have a **15-calendar-day UAT review window** to test functionality and raise defect tickets.
- If no blocking issues are logged within 15 days, the milestone is deemed formally accepted, authorizing the corresponding invoice tranche release.

2. Source Code, Intellectual Property & Architecture Transfer

- All custom schemas, PHP MVC engines, React front-end components, and integration scripts developed for Module 2 become the exclusive intellectual property of Popular Auto Spare & A/C Parts Co. W.L.L upon final settlement.
- Zero ongoing vendor lock-in; complete developer documentation and database schemas will be handed over.

3. Defect Classification & Warranty Period

- **Severity 1 (Critical):** System halt or procurement blocker • Fix within 4 business hours.
- **Severity 2 (High):** Workflow or approval degradation • Fix within 12 business hours.
- **Severity 3 (Standard):** Cosmetic or minor report enhancement • Fix in scheduled sprint.
- **Warranty:** 60 calendar days of comprehensive bug-fix warranty following production sign-off.

07

Section 7.0: Stakeholder Authorization & Digital Sign-Off Console

MULTI-PARTY RATIFICATION

By signing below, authorized representatives ratify the technical scope, 12-week deliverable timeline, dedicated engineering team cost matrix, and milestone payment tranches for Module 02:

Client Authorized Signatory Popular Auto Spare & A/C Parts Co. W.L.L Pending Ratification Signatory: Managing Director / General Manager	Solutions Architect Lead SaNDS Lab Middle East W.L.L Pending Ratification Signatory: Enterprise Solutions Architect	Project Management Advisor UniGlobal Business Solutions Pending Ratification Signatory: Strategic Project Management Consultant
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